

- Document Control
- System Name: Application
- Executive Summary
- This Technical Requirements Document outlines the technical implementation of the functional requirements for data processing and aggregation involving customer, order, ordersummary, and customeraggregatespend Delta tables.
- Technical Requirements

TR-CST-001

- Related Functional Requirement: FR-CST-001
- Objective: Read source CSV data from the specified volumes and load into Delta tables: customer and order.
- Target Cluster Configuration:
 - Cluster Name:
 - Runtime Version:
 - Node Type:
 - Driver Node:
 - Worker Nodes:
 - Autoscaling:
 - Auto Termination:
 - Libraries:
- Source Data Details:
 - customer: /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata
 - order: /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata
- Target Data Details:
 - Delta tables: customer, order
- Job Flow:
 1. Read CSV data from source paths
 2. Load data into respective Delta tables
- Transformations & Logic:
 - Map CSV columns to Delta table schema
- Error Handling & Logging:

- Log failed loads with reasons
- Capture and report schema mismatches

TR-CST-002

- Related Functional Requirement: FR-CST-002
- Objective: Apply schema for both customer and order tables.
- Target Cluster Configuration:
 - Cluster Name:
 - Runtime Version:
 - Node Type:
 - Driver Node:
 - Worker Nodes:
 - Autoscaling:
 - Auto Termination:
 - Libraries:
- Source Data Details:
 - customer data, order data
- Target Data Details:
 - customer: CustId, Name, EmailId, Region
 - order: OrderId, ItemName, PricePerUnit, Qty, Date, CustId
- Job Flow:

1. Apply schema on data load for both customer and order tables

- Transformations & Logic:
 - Enforce defined field types and order
- Error Handling & Logging:
 - Capture and report type mismatches or missing fields

TR-ORD-003

- Related Functional Requirement: FR-ORD-003
- Objective: Add "TotalAmount" column to the order table as PricePerUnit * Qty.

- Target Cluster Configuration:
 - Cluster Name:
 - Runtime Version:
 - Node Type:
 - Driver Node:
 - Worker Nodes:
 - Autoscaling:
 - Auto Termination:
 - Libraries:

 - Source Data Details:
 - order table with PricePerUnit and Qty columns

 - Target Data Details:
 - order table with additional TotalAmount column

 - Job Flow:
1. Process each order record
 2. Compute $\text{TotalAmount} = \text{PricePerUnit} * \text{Qty}$
 3. Store computed value in TotalAmount column
- Transformations & Logic:
 - Multiply numeric fields, handle nulls

 - Error Handling & Logging:
 - Log records with missing/invalid PricePerUnit or Qty

TR-CST-004

- Related Functional Requirement: FR-CST-004
- Objective: Cleanse data by removing Null and duplicate records from both tables.

- Target Cluster Configuration:
- Cluster Name:
- Runtime Version:
- Node Type:
- Driver Node:

- Worker Nodes:
- Autoscaling:
- Auto Termination:
- Libraries:
- Source Data Details:
- customer and order tables
- Target Data Details:
- Cleansed customer and order tables
- Job Flow:

1. Scan both tables for Null fields

2. Remove records with Nulls

3. Remove duplicate records

- Transformations & Logic:
- Conditional deletion, record matching
- Error Handling & Logging:
- Log records removed and reasons

TR-ORD-005

- Related Functional Requirement: FR-ORD-005
- Objective: Create "ordersummary" table if it does not exist in specified catalog and schema.
- Target Cluster Configuration:
- Cluster Name:
- Runtime Version:
- Node Type:
- Driver Node:
- Worker Nodes:
- Autoscaling:
- Auto Termination:
- Libraries:
- Source Data Details:

- Databricks catalog, schema

- Target Data Details:

- ordersummary table

- Job Flow:

1. Check for table existence

2. Create if not exists

- Transformations & Logic:

- Table DDL

- Error Handling & Logging:

- Log creation success or errors

TR-ORD-006

- Related Functional Requirement: FR-ORD-006

- Objective: Join customer and order data on "CustId" and load to SCD type 2 "ordersummary" table.

- Target Cluster Configuration:

- Cluster Name:

- Runtime Version:

- Node Type:

- Driver Node:

- Worker Nodes:

- Autoscaling:

- Auto Termination:

- Libraries:

- Source Data Details:

- customer and order tables

- Target Data Details:

- ordersummary SCD type 2 table

- Job Flow:

1. Join tables using CustId

2. Structure result per "ordersummary" schema

- Transformations & Logic:
- Merge logic
- Maintain changing customer attributes
- Error Handling & Logging:
- Log failed joins

TR-ORD-007

- Related Functional Requirement: FR-ORD-007
- Objective: Maintain "ordersummary" schema as CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date.
- Target Cluster Configuration:
- Cluster Name:
- Runtime Version:
- Node Type:
- Driver Node:
- Worker Nodes:
- Autoscaling:
- Auto Termination:
- Libraries:
- Source Data Details:
- Upstream data frames
- Target Data Details:
- ordersummary fields
- Job Flow:

1. Map data to schema

- Transformations & Logic:
- Enforce column ordering
- Error Handling & Logging:
- Log schema mismatches

TR-ORD-008

- Related Functional Requirement: FR-ORD-008
- Objective: Update "ordersummary" as SCD type 2. Old records set Inactive, new records Active, with StartDate, EndDate for history tracking.
- Target Cluster Configuration:
 - Cluster Name:
 - Runtime Version:
 - Node Type:
 - Driver Node:
 - Worker Nodes:
 - Autoscaling:
 - Auto Termination:
 - Libraries:
- Source Data Details:
 - customer updates
- Target Data Details:
 - ordersummary SCD type 2
- Job Flow:
 1. Monitor changes in customer
 2. Apply SCD type 2 logic in ordersummary
- Transformations & Logic:
 - Inactivate old records, insert new, update dates
- Error Handling & Logging:
 - Log changes and failures

TR-AGR-009

- Related Functional Requirement: FR-AGR-009
- Objective: Create "customeraggregatespend" table if not exists, with columns "Name", "TotalAmount", and "Date".

- Target Cluster Configuration:
- Cluster Name:
- Runtime Version:
- Node Type:
- Driver Node:
- Worker Nodes:
- Autoscaling:
- Auto Termination:
- Libraries:
- Source Data Details:
- Not applicable
- Target Data Details:
- customeraggregatespend table
- Job Flow:

1. Check if table exists

2. Create if needed

- Transformations & Logic:
- Table DDL
- Error Handling & Logging:
- Log creation attempts

TR-AGR-010

- Related Functional Requirement: FR-AGR-010
- Objective: Aggregate "TotalAmount" from "ordersummary", grouped by "Name" and "Date", and load into "customeraggregatespend".
- Target Cluster Configuration:
- Cluster Name:
- Runtime Version:
- Node Type:
- Driver Node:
- Worker Nodes:

- Autoscaling:
- Auto Termination:
- Libraries:
- Source Data Details:
- ordersummary
- Target Data Details:
- customeraggregatespend
- Job Flow:

1. Group ordersummary by Name, Date

2. Sum TotalAmount

3. Insert results to customeraggregatespend

- Transformations & Logic:
- Aggregation, grouping
- Error Handling & Logging:
- Log aggregation errors

TR-DLT-011

- Related Functional Requirement: FR-DLT-011
- Objective: Implement steps 1 through 10 with Delta Live Tables.
- Target Cluster Configuration:
- Cluster Name:
- Runtime Version:
- Node Type:
- Driver Node:
- Worker Nodes:
- Autoscaling:
- Auto Termination:
- Libraries:
- Source Data Details:
- All prior source datasets

- Target Data Details:

- All target tables

- Job Flow:

1. Script Delta Live Table pipelines

- Transformations & Logic:

- Build pipeline with defined transformations

- Error Handling & Logging:

- Integrate pipeline monitoring and logging